

FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION
SSC PART-I (CLASS 9) PRACTICE EXAMINATION
CHEMISTRY PAPER 25

Syllabus: Chapter 10, Acids, Bases and Salts (35 marks) + Revision: Chapters 6 and 8 (15 marks)

Total Marks: 50

Time Allowed: 2 hours 15 minutes

Version A

HIDAYAAT (Instructions)

1. **Saaray sawal** hal karnay hain. Is paper mein koi bhi **optional sawal nahin** hai.
2. Jawab apni notebook mein likhein. Har sawal ka number **bilkul waisa hi** likhein jaisa paper par hai, aur tarteeb se hal karein.
3. Section A mein sawal ka number likh kar sirf apnay option ka **letter** likhein, jaise **1. (b)**. Check kar lein ke jo letter aap ne likha hai, wohi option aap ne choose kiya hai.
4. Har equation **balance** karein. Dono taraf atoms gin kar check karein.
5. Har calculation mein formula aur working likhein. **Calculator use kar sakti hain**. Har jawab ke saath **unit** zaroor likhein.
6. Jahan **naam** maanga gaya ho wahan angrezi naam likhein. Naam ya unit na likhne par $\frac{1}{2}$ **mark** katega.
7. Har explanation aur calculation ke aakhir mein **ek mukammal jumla** likhein jo sawal ka jawab de.

SECTION A: MULTIPLE CHOICE QUESTIONS (10 × 1 = 10 MARKS)

#	Question	A	B	C	D	Answer
1	Which of the following cannot be classified as an Arrhenius acid?	HNO ₃	H ₂ CO ₃	CO ₂	H ₂ SO ₄	(A)(B)(C)(D)
2	Which of the following is a Bronsted-Lowry base ?	NH ₃	HCl	CH ₃ COOH	H ₃ O ⁺	(A)(B)(C)(D)
3	Milk of magnesia contains Mg(OH) ₂ . It is used as an antacid to neutralize excess stomach acid (hydrochloric acid, HCl). Which salt is formed?	MgSO ₄	MgCO ₃	MgCl ₂	MgO	(A)(B)(C)(D)
4	According to the Bronsted-Lowry theory, ammonia is a base because it	contains an OH group	donates a proton	produces H ⁺ ions	accepts a proton	(A)(B)(C)(D)
5	Which one of these is a weak base?	NaOH	Al(OH) ₃	KOH	Ca(OH) ₂	(A)(B)(C)(D)
6	According to Table 10.1, which property do acids and bases share ?	sour taste	turn blue litmus red	bitter taste	their aqueous solutions conduct electricity	(A)(B)(C)(D)
7	A strong base is a base that	completely dissociates in aqueous solution	dissolves in water	partially dissociates in aqueous solution	has a bitter taste	(A)(B)(C)(D)
8	For C _(s) + H ₂ O _(g) → CO _(g) + H _{2(g)} , ΔH° = +131.4 kJ. This reaction is	exothermic	endothermic	at equilibrium	neutral, with no heat change	(A)(B)(C)(D)
9	The mass of 0.75 mol of carbon dioxide, CO ₂ , is (C = 12, O = 16)	44 g	33 g	22 g	11 g	(A)(B)(C)(D)
10	How many moles are present in 54 g of water, H ₂ O? (H = 1, O = 16)	3 mol	0.33 mol	54 mol	1.5 mol	(A)(B)(C)(D)

SECTION B: SHORT ANSWER QUESTIONS (6 × 4 = 24 MARKS)

Chhay ke chhay sawal hal karein. Har sawal 4 marks ka hai.

2.	(i) Define Arrhenius acids and give two examples. (2) (ii) Why does HCl act as a strong acid? (1) (iii) Why does ammonia act as a weak base? (1)	4
3.	(i) State two limitations of the Arrhenius theory of acids and bases. (2) (ii) $\text{NH}_3 + \text{HNO}_3 \rightarrow \text{NH}_4\text{NO}_3$. Using the Bronsted-Lowry theory, explain why NH_3 is the base and HNO_3 is the acid in this reaction. (2)	4
4.	(i) $\text{HNO}_3 + \text{H}_2\text{O} \rightarrow \text{H}_3\text{O}^+ + \text{NO}_3^-$. Identify the Bronsted-Lowry acid and the Bronsted-Lowry base, giving a reason for each. (2) (ii) Write the equation for the ionization of (a) KOH and (b) NH_4OH in water, with charges and the correct arrow. Which of the two is the weak base? (2)	4
5.	Write a balanced chemical equation for each of these neutralization reactions: (i) ammonium hydroxide + nitric acid → ammonium nitrate + water (2) (ii) sulphuric acid + magnesium hydroxide → magnesium sulphate + water (2)	4
6.	Using Table 10.1 of your book, compare an acid and a base under each heading: (i) taste (ii) effect on blue litmus (iii) effect on red litmus (iv) effect on skin (4)	4
7.	The bond energies are: H-H = 436 kJ/mol, I-I = 151 kJ/mol, H-I = 299 kJ/mol. Calculate ΔH for the reaction $2\text{HI(g)} \rightarrow \text{H}_2\text{(g)} + \text{I}_2\text{(g)}$, showing the formula you use, and state whether the reaction is exothermic or endothermic, with a reason. (4)	4

SECTION C: LONG ANSWER QUESTIONS (2 × 8 = 16 MARKS)

Dono sawal hal karein. Har sawal 8 marks ka hai.

8.	Proton donors and proton acceptors. (i) Identify the Bronsted-Lowry acid and the Bronsted-Lowry base in each reaction, giving a reason for each: (4) (a) $\text{HCO}_3^- + \text{H}_2\text{O} \rightleftharpoons \text{CO}_3^{2-} + \text{H}_3\text{O}^+$ (b) $\text{HCl} + \text{HCO}_3^- \rightleftharpoons \text{H}_2\text{CO}_3 + \text{Cl}^-$ (ii) Using your answers to (a) and (b), explain why HCO_3^- can be described as amphoteric . (2) (iii) In $\text{HS}^- + \text{H}_2\text{O} \rightleftharpoons \text{S}^{2-} + \text{H}_3\text{O}^+$, is water a proton donor or a proton acceptor? Give a reason. (2)	8
9.	Zinc and hydrochloric acid. Zinc reacts with dilute hydrochloric acid: $\text{Zn} + 2\text{HCl} \rightarrow \text{ZnCl}_2 + \text{H}_2$ A student uses 6.5 g of zinc . (Zn = 65, H = 1, Cl = 35.5, $N_A = 6.02 \times 10^{23}$) (i) Calculate the number of moles of zinc used. (1) (ii) Calculate the number of moles, and the mass, of HCl needed to react with it completely. (2) (iii) Calculate the number of moles, and the mass, of hydrogen gas produced. (2) (iv) Calculate the number of hydrogen molecules produced. (1) (v) Calculate the mass of zinc chloride formed. (1) (vi) Write one concluding sentence that states the mass of hydrogen and of zinc chloride produced. (1)	8

END OF QUESTION PAPER

Chapters covered: Chapter 10 Acids, Bases and Salts (MCQs 1 to 7, Questions 2 to 6 and 8) | Revision: Chapter 8 Energetics (MCQ 8, Question 7), Chapter 6 Stoichiometry (MCQs 9 and 10, Question 9).

Student's Name: _____

Date: _____

Marks Obtained: _____ / 50